HANUMANT VITTHAL KSHIRSAGAR

B.Tech. - Computer Engineering

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PROFESSIONAL SUMMARY

Final-year Computer Engineering student with strong skills in Python, Java, C++, SQL, AI/ML, Deep Learning, Data Analysis, and Full-Stack Development. Experienced in applying machine learning, computer vision, and data-processing techniques through academic projects and an AI/ML internship at SAC, ISRO, focusing on satellite image processing. Seeking an entry-level opportunity to apply technical and problem-solving skills, contribute to real-world projects, and grow as a technology professional while creating value for the organization.

KEY EXPERTISE

Artificial Intelligence | Machine Learning | Deep Learning | Computer Vision | Data Analysis | Data Processing | Python Programming | Java Programming | C++ Programming | SQL | Data Visualization | Full-Stack Development | Web Development | Satellite Image Processing | Problem Solving | IoT

EDUCATION

AISSMS Institute of Information Technology 2023 - 2027

B.Tech. - Computer Engineering | CGPA: **9.09** / **10**

Dada Patil College, karjat 2022

12th | MSBSHSE | Percentage: **82.17** / **100**

Shree Chhatrapati Shivaji Vidyalaya, Veet, Karmala 2020

10th | MSBSHSE | Percentage: **94.80** / **100**

INTERNSHIPS

Space Applications Centre (SAC), ISRO | Aviation / Aerospace 01 Jun, 2026 - 28 Aug, 2026

AI/ML Intern

Key Skills:

Python | Data Preprocessing, Data Cleaning ,Exploratory Data Analysis ,Data Visualization | Image Deblurring | Remote Sensing | Model Building, Model Training ,Model Evaluation | Satellite Image Processing | Artificial Intelligence | Deep Learning | Image Preprocessing | Computer Vision | Blind Deconvolution | Image Restoration

Worked as an AI/ML Intern at the Space Applications Centre (SAC), ISRO, Ahmedabad, focusing on satellite image data preprocessing and image restoration. Worked with Cartosat-2E high-resolution panchromatic satellite imagery, including PSF/MTF estimation, blind deconvolution, and deep-learning-based image deblurring. Explored image preprocessing, restoration techniques, and evaluation of image quality using quantitative image-quality metrics. Gained practical experience in satellite image processing, computer vision, deep learning, and remote sensing applications.

PROJECTS

EcoMind – Environmental Intelligence & Air Quality Monitoring Platform

Mentor: Mr.Sugat Pawar | **Team Size:** 3

Key Skills:

Python | Flask | IoT | Web Development | Artificial Intelligence | Machine Learning | Data Processing | Environmental Data Analysis | Real-Time Data Visualization | Firebase

Developed EcoMind, an Environmental Intelligence Platform for real-time monitoring and analysis of air quality and environmental conditions using an ESP32-based IoT system. Integrated multiple sensors including PMS5003, MQ-7, MiCS-2714, MQ-131, and DHT22 to collect particulate matter, gas concentration, temperature, and humidity data. Implemented a Flask-based backend for data collection, processing, storage, and analysis, with AI/ML-based risk prediction to classify environmental conditions into Safe, Moderate, and Risky categories. Developed an interactive web dashboard using HTML, CSS, and JavaScript for real-time visualization, alerts, and administrative monitoring. Integrated Firebase for data management and designed the platform to support continuous environmental monitoring and data-driven decision-making.

PUBLICATIONS / RESEARCH / WHITE PAPERS

An IoT-Based Machine Learning Framework for Environmental Quality Assessment and Human Living Comfort Prediction

Conference Paper/Research Paper | IEEE I2ITCON 2026 | **Mentor:** Mr.Sugat Pawar | **No. of Authors:** 4

Key Skills:

Artificial Intelligence | Machine Learning | Environmental Monitoring | Sensor Data Processing | Data Analysis | Data Visualization | Python | ESP32 | Predictive Analytics | Risk Assessment | FastAPI | Firebase | IoT

The paper presents an IoT-based machine learning framework for real-time environmental quality assessment and human living comfort prediction. Existing environmental monitoring systems often rely on expensive fixed stations, have limited flexibility and coverage, and primarily focus on pollution levels without considering human comfort and exposure risk. The proposed framework addresses these limitations by integrating an ESP32-based sensing system with environmental sensors, cloud-based data management, and machine learning-based classification.

The system collects PM2.5, NO2, CO, O3, temperature, and humidity data through sensors connected to an ESP32 and transmits the data to Firebase for storage and analysis. A WHO-guided weighted risk scoring and labeling approach is used to generate Safe, Moderate, and Risky environmental condition classes. The study evaluates Logistic Regression, Random Forest, SVM, and KNN models for classification. Random Forest achieved the highest reported performance, with 99.22% accuracy and a 99.24% F1-score. A sensor-failure ablation study was also conducted to evaluate the fault tolerance of the ML-based sensor-fusion approach. The work demonstrates the integration of IoT, data processing, machine learning, cloud computing, and environmental analytics for scalable real-time monitoring.

ACHIEVEMENTS

- Selected for AI/ML Internship at Space Applications Centre (SAC), ISRO, Ahmedabad (2026)
- Selected for Paid Internship at Centre for Artificial Intelligence and Robotics (CAIR), DRDO, Bengaluru (2026)
- Design Patent — Farm Monitoring AI-Machine, Registered Design No. 476772-001, Indian Patent Office (February 2026)
- mart India Hackathon — Top 10 among 160+ teams
- National-level Techthon 3.0 — Runner-up, Environment Domain

ASSESSMENTS / CERTIFICATIONS

Mobile App Development and Mastering UI with HTML and CSS

Provider: Maharashtra Knowledge Corporation Limited (MKCL)
Aggregate: 92 / 100
Subjects: Mobile App Development, HTML, CSS, UI Design (92 / 100)

Key Skills: Mobile App Development | HTML | CSS | UI Development | Web Development

Successfully completed a 120-hour iLike credit course in Mobile App Development and Mastering UI with HTML and CSS, including learning, assessment, and examination. The course provided practical knowledge of mobile application development, HTML, CSS, and user interface development. Secured 92 out of 100 marks and was awarded 4 academic credits.

CO-CURRICULAR ACTIVITIES

- Technical Head, National Service Scheme (NSS)
- Discipline & Security Coordinator, Computer Engineering Students Association (CESA)
- Member, ACM Student Chapte

EXTRA CURRICULAR ACTIVITIES

- District-level Kho-Kho Player
- District-level Kabaddi Player
- College Kabaddi Team Member
- National Service Scheme (NSS) — Active Volunteer
- Participation in Hackathons and Ideathons

PERSONAL INTERESTS / HOBBIES

- Exploring Emerging Technologies
- Coding and Software Development
- Research and Innovation
- Sports and Fitness
- Team Activities

PERSONAL DETAILS

Gender: Male
Marital Status: Single
Current Address: near Sutar hospital ,Gujrati colony ,Kothrud, Pune, Pune, Maharashtra, India - 411038

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Date of Birth: 03 Jul, 2004
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